Peter Laszcz

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I am currently in my final year of an integrated masters in Mathematics and Computer Science at the University of Birmingham. I have developed skills in software development and mathematics through both employment and educational experience. I supplement my academic interests through personal and independent research and projects, and I wish to pursue a career in research with computer science and applied mathematics after I graduate.

Work Experience

Scientific Software Placement with Science and Technology Facilities Council (STFC) (May-Sept 2026)

- Imaging software development position at the ISIS Neutron and Muon Source, Rutherford Appleton Laboratory.
- Produced data analysis, processing, and visualisation software for a novel neutron imaging method (Spectroscopic Neutron Imaging).
- Collaborated internally to integrate the code within the facility's long-term processing codebase, and externally on an international level to bring the program to other neutron imaging facilities.
- Documented the developments encapsulated within the program in a paper to be submitted for first-author publication.
- Received training in CI/CD, project management, and documentation systems.
- Gave seminars and technical presentations on the project, as well as producing outreach content for the facility.
- Contributed sections to paper output (joint authorship), and content for presentation at conferences across Europe.

Open-Source Developer for the European Molecular Biology Laboratory (May-Sept 2025)

- Stipend received from Google (via the Python Software Foundation) through the Google Summer of Code program to develop features of Ilastik (imaging and image-processing software) under the mentorship of Research Software Engineers at the European Molecular Biology Laboratory (EMBL).
- Features were requested by corresponding institutions to improve data pipeline efficiency through metadata preservation.
- Operated within an international group with Agile methodology and Test-Driven Development principles.
- Delivered project progress updates at weekly team meetings.
- Merged requested features by successful completion of the project for stable release.

Research Associate at the University of Birmingham (Oct-Dec 2024)

- Programmer and mathematical modelling research associate for ARME, an EPSRC-funded project investigating musician synchronisation in ensemble performance.
- Responsibilities included: researching novel synchronisation model literature, implementing models computationally, performing model analysis with experimental data, managing the source code via Version Control Systems, providing code and model documentation, attending weekly team meetings to present updates and results.
- Findings resulted in paper co-authorship (currently in review, PLOS Computational Biology).

Research Intern at the University of Birmingham Positron Imaging Centre (Jun-Aug 2024)

- Conducted imaging and particle-tracking data analysis on reactor systems in the context of plastics pyrolysis.
- Utilised Positron Emission Particle Tracking data to model fluid dynamics within the system, as well as analysing radiographs via Convolutional Neural Networks to locate and quantitively measure gas structures.
- Deployed and maintained novel software on the university's HPC system, as well as developing features of existing in-house software.
- Discussed the aims, progress, and impact of the research project with academics from the university and abroad at conferences.
- Yielded a 54% improvement in structure measurement over classical techniques, as well as enabling the analysis of additional bed structures.

Education

M.Sci Mathematics and Computer Science at the University of Birmingham (Sept 2023 – Jun 2027)

Graduating 2027. Predicted First Class Honours. Masters project on imaging large magnetogram datasets in collaboration with the British Geological Survey.

A-Levels at the Cardinal Vaughan Memorial School (Sept 2021 – Jun 2023)

Mathematics A*, Physics A*, Chemistry A, Further Mathematics B

Volunteering Positions

Secretary at University of Birmingham Disabled Students' Association (SANDAM) (2025-2026)

- Took minutes for meetings and processed administration data, including managing room allocations to ensure meetings and events could go ahead.
- Corresponded with student union and university administration, and collaborated with faculties, to raise awareness and promote accessibility for disabled students.
- Collaborations lead to successful delivery of disability employability and awareness seminar series.

Treasurer at Ruislip SVP and University of Birmingham G&S Society (2021-2022; 2024-2026)

- Collected & processed financial and administration data for two nonprofit societies.
- Collated receipts and invoices to produce and present reports and determine methods of managing costs.
- Worked with event organisers to calculate budgets, apply for grants, and manage costs.
- Treasurer work with student societies resulted in excess increase of 175% across tenure.

Young Scout Leader at 4th Eastcote Scout Troop (2021-2022)

- Planned, organised, and ran a weekly activity schedule with a team of leaders for a scout patrol.
- Activities included external excursions and moderate risk activities such as using axes and knives, building fires, climbing.
- The position developed my confidence in project and risk management and bolstered my skills in leadership, teamwork, and communication.

Non-Academic Achievements and Interests

- **Programming:** Proficiency in multiple programming (C, C++, Haskell, Java, Python, MATLAB) and markup (LaTeX, Markdown) languages. Project interests include Computer Vision, cryptography, and High Performance Computing. Experience in hackathons (BirmingHack 2025; BirmingHack 2026 (second place); University of Birmingham Secure Compute Hackathon 2026 (produced whitepaper on implementation of Trusted Research Environments in High-Performance Computing for university executives)).
- **Institute of Mathematics and its Applications (IMA):** Associate Member.
- **Mathematics Engagement:** Previous engagement work has included mathematics aid at Peer Assisted Study Sessions, and secondary-level demonstrations with the IMA at a national science and technology fair.
- **University Societies:** Gilbert & Sullivan Society, Music Society, Maths Society, Computer Science Society, Student Disability Association.
- **Music:** Violin Grade 8, 13 years choral experience (recent performances include a collaboration with the BBC Singers, and at St. Paul's Cathedral, London), and 10 years orchestral experience (most recently performing in a tour in Sicily).

Referees

Dr. Galane J. Luo (Personal Academic Tutor)
Mathematics Lecturer at the University of Birmingham
Email: g.j.luo@bham.ac.uk

Robert Scott Young (Previous Line Manager)
Postdoctoral researcher at the ISIS Neutron and Muon Source
Email: scott.young@stfc.ac.uk